Lecture 3: The Diamond Model

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Outline

- Critique of the Solow-Swan model
- Environments of the Diamond model
  - Individuals
  - Firms
  - Markets
  - Timing of events
- Competitive equilibruim
  - Firms' profit maximisation
  - Individuals' utility maximisation

Critique of the Solow-Swan Model (1 of 2)

- Merits
  - The model identifies several important sources of growth: capital accumulation (through saving and investment), and technological progress.
  - It leads to a number of testable prediction that stand up quite well in empirical testing.
  - It proposed a basic framework for neoclassical growth model.
  - It is one of the most widely applied models in economics, instructing a lot of empirical studies on the determinants of growth (e.g. growth accounting, growth regressions).

Critique of the Solow-Swan Model (2 of 2)

- Criticism
  - The model takes the saving rates as exogenous and fixed.
  - Treatment of technology is highly incomplete.
    - The model does not identify what the "effectiveness of labour" is.
    - Technological progress, the only force giving rise to per-capita growth in long term, is exogenous.
  - The model does not incorporate human capital, which is also viewed as important for growth.

What is Next?

- Extending the Solow model
- Endogenizing the saving rate: The Diamond model and the Ramsey model
  - Both model continue to take the growth rates of L and A as given, but investigate the determinants of saving and investment from decision at the microeconomic level.
  - The two model present two basic framework for modeling the macroeconomy.
    - The Diamond model (Diamond(1965)): OLG framework
    - The Ramsey model: Representative agent framework
- New growth theory
  - Identify A and endogenize its evolution
  - Incorporate human capital and model its accumulation.

Motivation of OLG Model

- Neoclassical growth model relies on the representative household.
- Diamond model allows for the arrival of new households into the economy: realistic (takes account of life cycle), but also introduces a range of new economic interactions. Decisions by older generations affect the prices faced by younger generations.

The Baseline OLG model

- The discussion of the Diamond model gives you an illustration of the neoclassical approach.
- **Time** is discrete and runs to infinity, $t = 0, 1, 2, ...$
- **Individuals**:
  - The economy is populated with 2-period lived individuals:
    - Those born at t lives at t and $t + 1$ only.
  - In period t, L_t new individuals are born and L_{t-1} old individuals are dying (two generations overlap). Assume population grows at rate n:

$$L_t = (1 + n)L_{t-1} \tag{1}$$

  - The initial old refers to the generation who are already in their old age when time starts ($t = 0$). All other generations are called future generations, who live for 2 periods.

Utility (1 of 2)

- **Preference**: The lifetime utility of an individual born at t is given by:

$$U(c_{1t}, c_{2t+1}) = u(c_{1t}) + \beta u(c_{2t+1}) \tag{2}$$

  - where c_{1t} and c_{2t+1} denote the consumption when young and when old (utility of an initial old is simply $u(c_2, 0)$)
  - U is additively separable.
  - $\beta\,(0 < \beta < 1)$ is the discount factor.
  - General assumptions on the periodic utility function u:
    - u is increasing ($u'(c) > 0$)
    - u is concave ($u''(c) < 0$)
    - u satisfies the Inada condition to ensure positive consumption in equilibrium ($\lim_{c \to 0} u'(c) = +\infty$)

Why the Inada condition rules out zero consumption? (1 of 2)

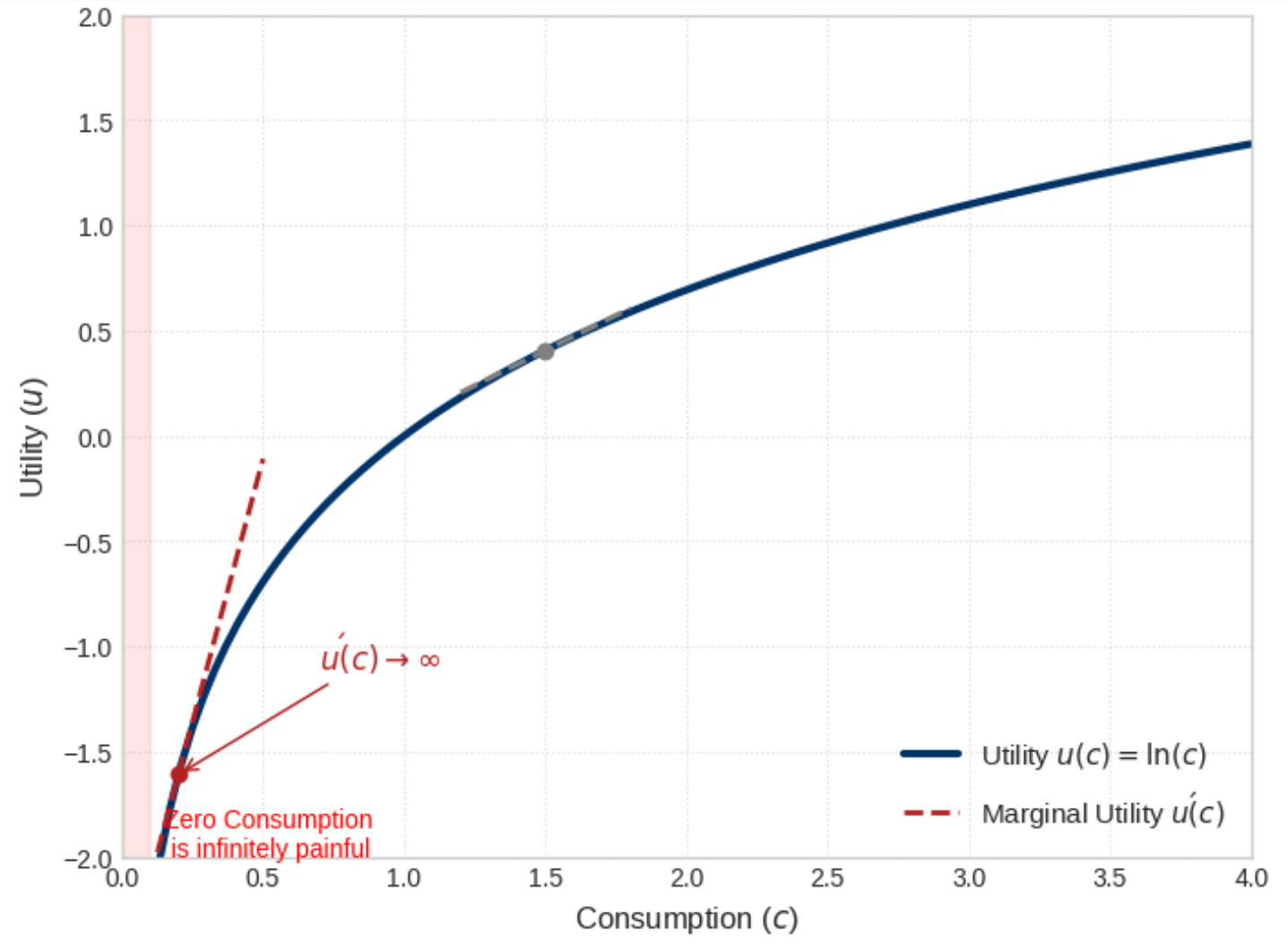
1. **The Condition** Recall the Inada condition:

$$\lim_{c \to 0} u'(c) = +\infty$$

2. **The Economic Intuition**

   - As $c \to 0$, the marginal gain from consuming just a little bit more becomes infinite.
   - Giving up a tiny amount of future consumption to avoid starvation today yields an enormous utility gain.

The Inada Condition: Why $c = 0$ is never optimal?



Why the Inada condition rules out zero consumption? (2 of 2)

3. The Euler Equation Proof At an optimum:

$$\underbrace{u'(c_1)}_{\text{Marginal Benefit}} = \beta(1+r) \underbrace{u'(c_2)}_{\text{Marginal Cost}}$$

If $c_1 = 0$, LHS goes to ∞. The equality cannot hold (unless $c_2 = 0$ too, which is impossible with positive income).

Key Takeaway

The Inada condition ensures an interior solution ($c_t > 0$). You never hit the corner constraint.

What Exactly is a "Corner Solution"? (1 of 2)

Definition

A corner solution occurs when the optimal choice lies on the boundary of the feasible set, rather than in the middle.

Examples in OLG:

- $c_1 = 0$: Starve when young (consume only later).
- $s = 0$: Save nothing (hand-to-mouth).
- $c_2 = 0$: Consume nothing when old.

What Exactly is a "Corner Solution"? (2 of 2)

Interior Solution:

Highest indifference curve is tangent to the budget line.

$$MRS_{c_1,c_2} = 1 + r$$

Calculus: FOC holds with equality.

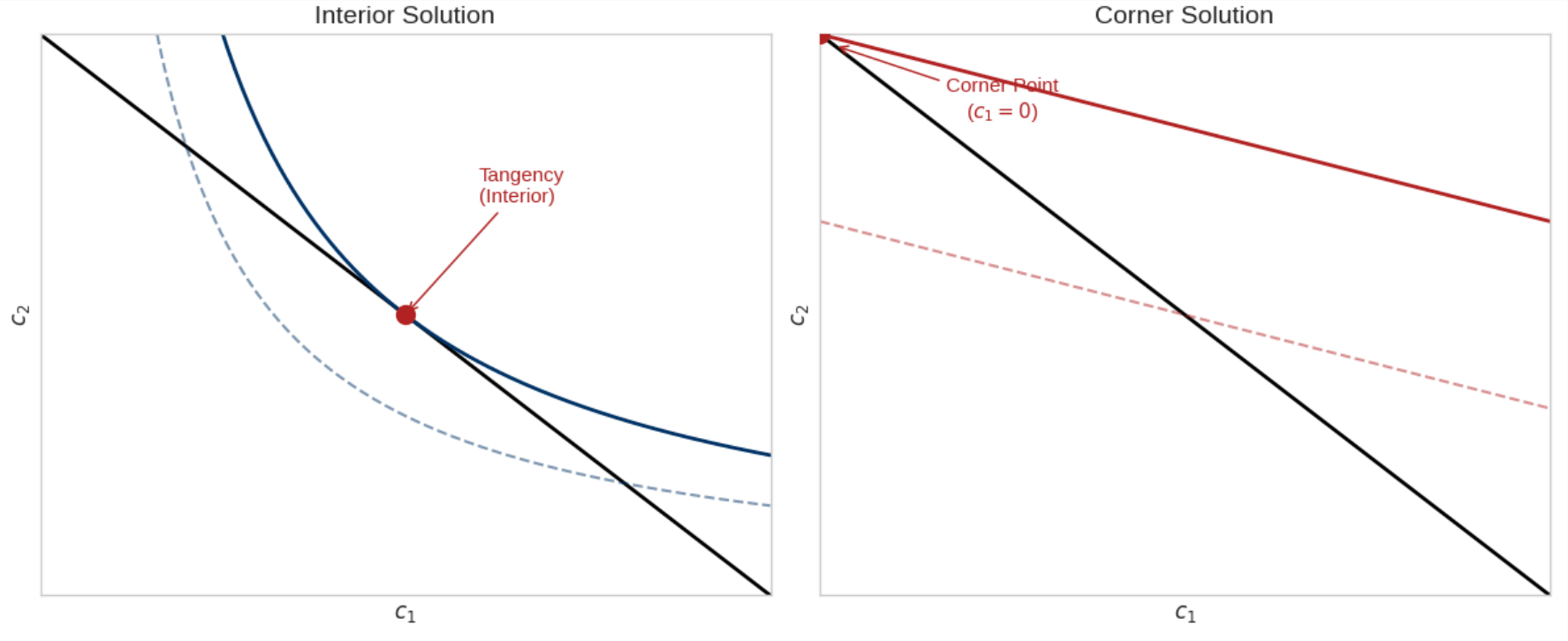
Corner Solution

Highest attainable indifference curve hits the endpoint.

$$MRS_{c_1,c_2} > 1 + r \quad (\text{if } c_1 = 0)$$

Calculus: FOC fails; inequality holds.

Interior Solution vs. Corner Solution



What if we did NOT assume Inada? (1 of 2)

The Hypothesis: Suppose marginal utility at zero is **finite**.

$$u'(0) < \infty$$

The Consequence: It becomes possible that the return to saving dominates the pain of starving:

$$\underbrace{u'(0)}_{\text{Pain of Hunger}} < \underbrace{\beta(1+r)u'(c_2)}_{\text{Benefit of Saving}}$$

What if we did NOT assume Inada? (2 of 2)

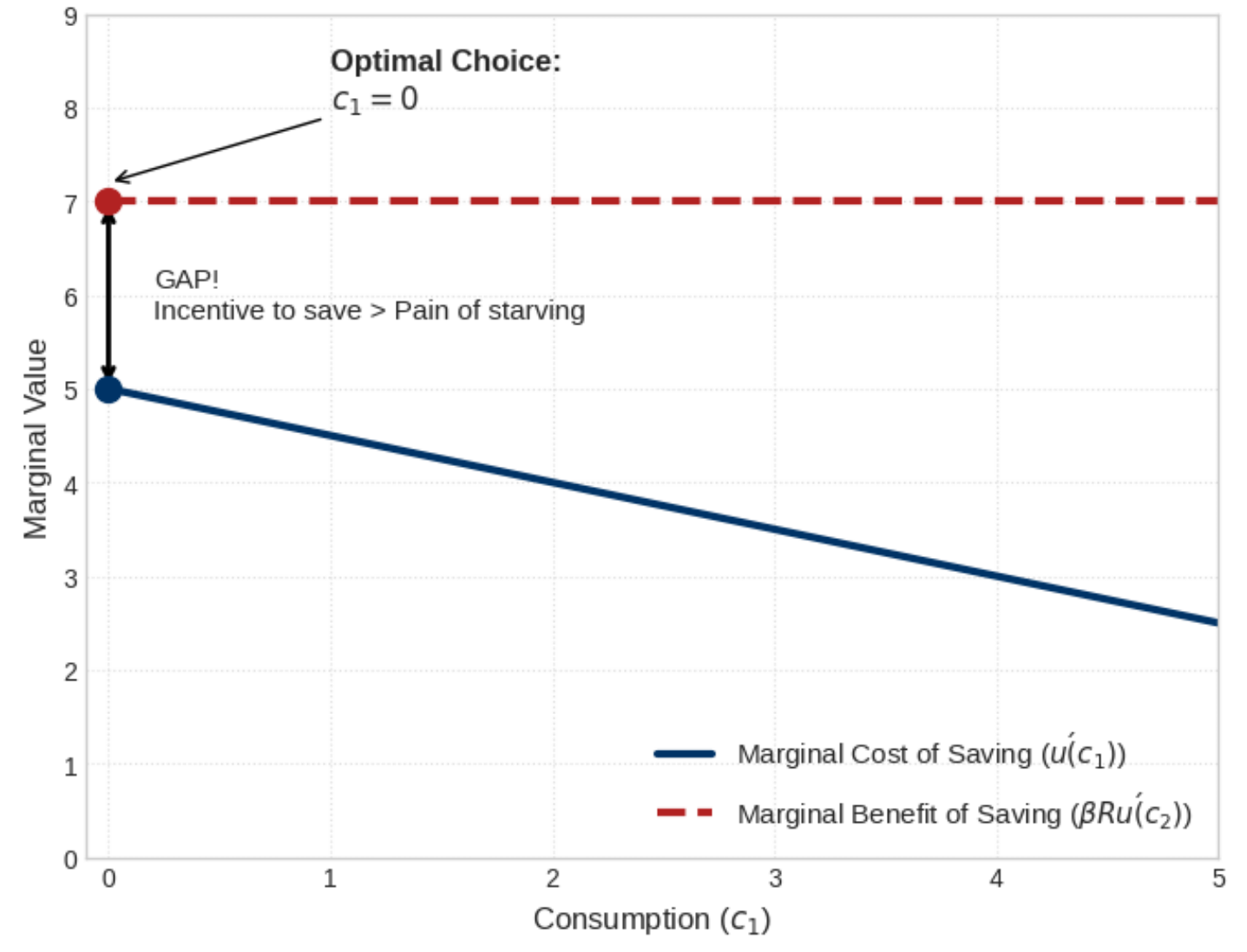
The Result ($c_1 = 0$):

- The optimal choice is to consume **nothing** today.
- "Extreme Specialization": Starve when young, feast when old.

Why this is "Pathological"

- Models without Inada conditions allow for corner solutions.
- That defy biological reality and create unstable "bang-bang" equilibria.

What if Inada fails? (Finite Marginal Utility)



Connection to Growth Models (notice something subtle)

In Solow, Inada on production ensures:

- Poor countries grow fast.
- Capital never collapses to zero.

In OLG, Inada on utility ensures:

- Consumption never collapses to zero.
- Saving decisions are interior.
- Euler equations characterize behavior.

Overall, Inada conditions are technical assumptions with powerful equilibrium implications.

Summary for Inada Condition

Inada condition on utility implies:

1. Marginal utility is infinite at zero consumption.
2. Zero consumption cannot satisfy the first-order condition.
3. Therefore optimal consumption must be strictly positive.
4. This guarantees interior solutions.
5. Interior solutions allow smooth equilibrium characterization.

Utility (2 of 2)

- To make some tight predictions, we need to put more structure on utility and production function (next).
- A widely used form u takes is the constant relative risk aversion (CRRA) utility function

$$u(c) = \frac{c^{1-\theta} - 1}{1 - \theta} = ln(c) \text{ if } \theta = 1 \tag{3}$$

  - Proof see Appendix 3
  - where $\theta > 0$ is the coefficient of relative risk aversion ($\frac{-cu''(\theta)}{u'(\theta)} = \theta$), the higher θ is, the more risk averse individuals are. It can be shown that $\frac{1}{\theta}$ is the elasticity of substitution between consumption in the two periods (see Appendix 4).

Endowment

- Each young individual is endowed with 1 unit of labour, which is supplied inelastically to firms.
- There is some initial capital stock K_0 owned equally by all initial old individuals.

Firms: Production Function

- In each period, firms hire labour and rent capital from individuals to produce output, sell the output in good market, and pay labour and capital. The CRS production function

$$Y_t = F(K_t, A_t L_t^D) \tag{4}$$

  - The intensive form of production: $y_t = f(k_t)$, where $k_t \equiv \frac{K_t}{A_t L_t^D}$. Assume that $f'(k) > 0, f''(k) < 0$ and $\lim_{k \to 0} f'(k) = +\infty$.
  - A_t grows at exogenous rate g.
  - There is full depreciation of capital.
  - With CRS, the number of firms can be normalized to 1.

Why CRS $\Rightarrow$ Zero Profits in Competitive Equilibrium? (1 of 3)

Step 1: Profit Maximization under Perfect Competition

Firms take (r, w) as given and choose K, L.

First-order conditions:

$$r = F_K(K, AL), \qquad w = \frac{\partial F}{\partial L} = F_{AL}(K, AL) \cdot A$$

(Note: because the second argument is AL, the marginal product w.r.t. L includes the A term.)

Why CRS $\Rightarrow$ Zero Profits in Competitive Equilibrium? (2 of 3)

Step 2: Apply Euler's Theorem (CRS)

If F is homogeneous of degree 1 (CRS), then:[1]

$$F(K, AL) = F_K(K, AL) \cdot K + F_{AL}(K, AL) \cdot (AL).$$

Using the FOCs:

$$F(K, AL) = rK + wL.$$

[1] See Appendix 5 for derivation.

Why CRS $\Rightarrow$ Zero Profits in Competitive Equilibrium? (3 of 3)

Step 3: Profits

$$\pi = F(K, AL) - rK - wL = 0.$$

Economic Intuition

With CRS, scaling the firm does not change average productivity. Under perfect competition, prices are bid so that firms earn zero pure profits.

Why factor payments exhaust output (no leftover)? (1 of 2)

Exhaustion Result:

From Euler's theorem under CRS:

$$Y = rK + wL.$$

Interpretation:

All output is paid out as income to factors:

- Capital owners receive rK
- Workers receive wL

There is no residual "profit wedge" in the competitive CRS case.

Why factor payments exhaust output (no leftover)? (2 of 2)

Why This Matters in Macroeconomics

This gives clean national-income accounting:

$$\text{Labor share} = \frac{wL}{Y}, \qquad \text{Capital share} = \frac{rK}{Y}.$$

Under Cobb–Douglas production, these shares often align closely with empirical factor shares (refer to Solow-Swan model).

Why CRS $\Rightarrow$ Clean Mapping from Marginal Products to Prices? (1 of 2)

In Competitive Markets:

- If a firm hires one more unit of capital at rental rate r, it compares r to the extra output produced: F_K.
- Optimality implies:

$$r = F_K.$$

Similarly for labor, the wage equals the marginal product of labor (adjusted for A if labor is effective labor):

$$w = MPL.$$

Direct Pricing Rule:

$$\boxed{r = MPK, \qquad w = MPL.}$$

Why CRS $\Rightarrow$ Clean Mapping from Marginal Products to Prices? (2 of 2)

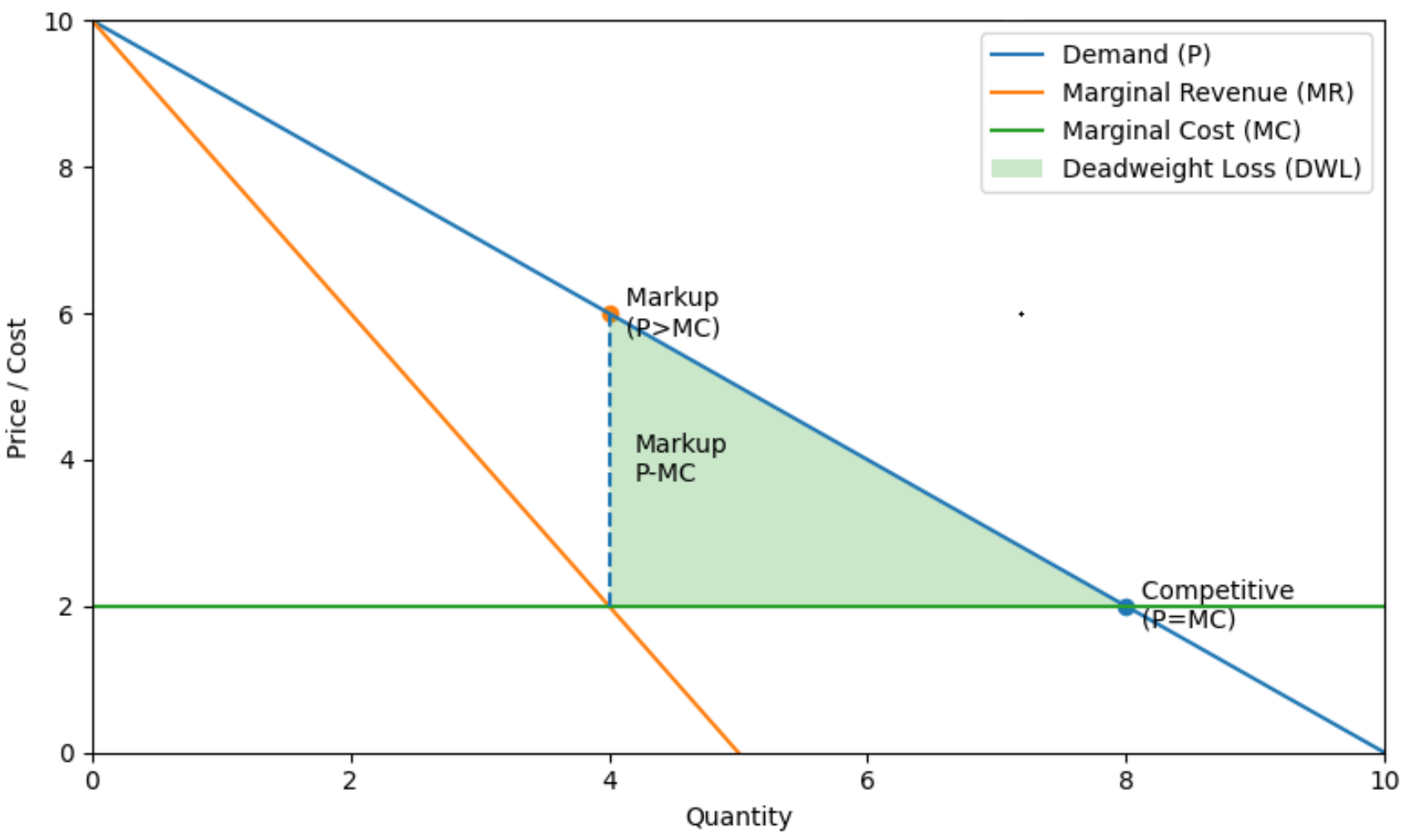
Why this is "Clean":

- It does not depend on firm size.
- It does not require bargaining.
- It does not require markups or market power.

If CRS or Perfect Competition Fails:

If CRS fails (e.g., increasing returns) or markets are not competitive (markups), then w and r need not equal marginal products, and profits/markups appear.

Competitive pricing vs. Markup pricing



Competitive pricing vs. Markup pricing: Key Intuition

- **Competitive equilibrium:** Firms are price takers $\rightarrow$ choose output where
$$P = MC \quad \Rightarrow \quad Q^C.$$
- **Markup (market power):** Firm chooses quantity where
$$MR = MC,$$
then charges price from the demand curve $\rightarrow$
$$P^M > MC \quad \text{at a lower quantity } Q^M.$$
- The shaded area is the **deadweight loss (DWL)**: mutually beneficial trades between Q^M and Q^C that do not occur because price is pushed above marginal cost.

Markets and Initial Conditions

- Markets are perfectly competitive.
- Denote the real interest rate (capital rental rate) and the real wage rate per unit of effective labour in period t as r_t and w_t, respectively.
- **Initial conditions**: A_0, L_{t-1} (number of initial old), K_0 are given.

In any given period t

- **Timing** of events in period t
  - Young individuals are born with labour endowments.
  - Firms hire labour from young individuals and rent capital from old individuals to produce output. Capital and labour are paid.
  - Old individuals consume capital income and existing wealth, then die and exit the economy.
  - Young individuals divide labour income between first-period consumption and saving to period $t+1$ which form new capital for period $t+1$ production.

The Competitive Equilibrium (Solution of the Model)

- **Firm's profit maximisation**
- Problem: The representative firm chooses capital demand and labour demand to maximize profits, taking interest rate and wages as given

$$\max_{K_t, L_t^D} F(K_t, A_t L_t^D) - r_t K_t - w_t A_t L_t^D$$

Recall that $F(K_t, A_t L_t^D) = A_t L_t^D f(k_t)$, $K_t = A_t L_t^D k_t$, thus

$$\max_{K_t, L_t^D} A_t L_t^D [f(k_t) - r_t k_t - w_t]$$

FOCs (for choice variables)

$$w.r.t\ k_t:\ r_t = f'(k_t) \tag{5}$$

$$w.r.t\ L_t^D:\ w_t = f(k_t) - f'(k_t) k_t \tag{6}$$

Firm's Problem (Cont'd)

- In fact, $f'(k_t)$ is the marginal product of capital ($MPK \equiv \frac{\partial F(K_t, A_t L_t^D)}{\partial K_t}$), and $f(k_t) - f'(k_t)k_t$ is the marginal product of effective labour ($\frac{\partial F(K_t, A_t L_t^D)}{\partial (A_t L_t^D)}$) (see Appendix 8 for proof)
- Firms' profit maximisation implies that factors of production earn their marginal products – a property of competitive markets.
- Notice that firms earn zero economic profits – a property of competitive markets.
- CRS can greatly simplifies firms' behavior. No need to worry about firm entry/exit, ownership of firms and dividends policy.
- Empirically, aggregate production function is close to CRS.

Individual's Problem (1 of 6)

- **Individual's utility maximisation**
- Situations faced by individuals:
  - The initial old simply rent the endowed capital to firms and consume the rental income and non-depreciated capital.
  - All future generations face the same situations: In the first period of life, supply labour and divide labour income between first-period consumption and saving (by holding capital). In the second period, simply consume the saving and any interests earned.

Individual's Problem (2 of 6)

- The only decision individuals need to make is for young individual to decide how much to consume today and how much to save for old ages, to maximize lifetime utility subject to budget constraints, taking prices (rational expectation applies) and A_t as given:

$$\max_{s_t} U(c_{1t}, c_{2t+1}) = u(c_{1t}) + \beta u(c_{2t+1})$$

s.t.

$$c_{1t} + s_t = w_t A_t \tag{7}$$

$$c_{2t+1} = (1 + r_{t+1}) s_t \tag{8}$$

  - where Eqs.(7) and (8) are the individual's budget constraints when young and when old. Note that the wage rate per worker or per unit of labour is $w_t A_t$.
  - FOC: $\frac{\partial U(c_{1t}, c_{2t+1})}{\partial s_t} = 0$

Individual's Problem (3 of 6)

$$\frac{\partial U(c_{1t}, c_{2t+1})}{\partial s_t} = U_1(c_{1t}, c_{2t+1})\frac{\partial c_{1t}}{\partial s_t} + U_2(c_{1t}, c_{2t+1})\frac{\partial c_{2t+1}}{\partial s_t} = 0$$
$$= u'(c_{1t})(-1) + \beta u'(c_{2t+1})(1 + r_{t+1}) = 0$$

Therefore,

$$\frac{u'(c_{1t})}{\beta u'(c_{2t+1})} = 1 + r_{t+1} \tag{9}$$

- Eq.(9) is called the consumption Euler equation, a necessary condition for individual's intertemporal consumption choice. Notice that

$$\frac{u'(c_{1t})}{\beta u'(c_{2t+1})} = \frac{U_1(c_{1t}, c_{2t+1})}{U_2(c_{1t}, c_{2t+1})} = MRS_{c_{1t}, c_{2t+1}}$$

- Hence, Eq.(9) states that to maximise the consumer lifetime utility, the marginal rate of substitution between current and further consumption must equal to the gross interest rate.

Individual's Problem (4 of 6)

- The second order condition (SOC) is given by:

$$\frac{\partial^2 U}{\partial s_t^2} = u''(c_{1t}) + \beta u''(c_{2t+1})(1 + r_{t+1})^2 < 0$$

  - Hence, there is a unique s_t that solves the individual's problem.
  - In summary, combining the Euler equation (9) and the budget constraints (7) and (8) gives us the equation that determines the unique solution s_t:

$$\frac{u'(w_t A_t - s_t)}{\beta u'[(1 + r_{t+1})s_t]} = 1 + r_{t+1} \tag{10}$$

  - Eq.(10) determines the saving s_t as a function of r_{t+1}, w_t, A_t and parameters, which the individuals takes as given. c_{1t} amd c_{2t+1} are then determined by Eqs.(7) and (8), i.e. the individual's problem is solved!

Individual's Problem (5 of 6)

- Another way to solve the individual utility maximisation problem is to solve a constrained maximisation problem.

$$\max_{c_{1t}, c_{2t+1}} U(c_{1t}, c_{2t+1}) = u(c_{1t}) + \beta u(c_{2t+1})$$
$$s.t. \quad c_{1t} + \frac{1}{1 + r_{t+1}} c_{2t+1} = A_t w_t \tag{11}$$

  - where Eq.(11) is an individual's lifetime budget constraint. It is obtained from the periodic budget constraints (7) and (8).
  - Notice the similarity between this formulation and a standard utility maximisation problem in microeconomics.

$$\max_{x_1, x_2} U(x_1, x_2) \quad s.t. \quad p_1 x_1 + p_2 x_2 = w$$

  - Here the two goods are first-period consumption and second-period consumption. The price for second-period consumption in terms of first-period consumption is $\frac{1}{(1+r_{t+1})}$.

Individual's Problem (6 of 6)

- Recall that FOC for the standard utility maximisation problem is given by:

$$\frac{U_1(x_1, x_2)}{U_2(x_1, x_2)} = \frac{p_1}{p_2}$$

- The above equation states that marginal rate of substitution between the two goods is equal to their price ratio. The FOC for our problem is given by (see Appendix 9 for proof):

$$\frac{U_1(c_{1t}, c_{2t+1})}{U_2(c_{1t}, c_{2t+1})} = 1 + r_{t+1}$$

- Notice that this equation is consistent with the standard form above: ($x_1 = c_{1t}$, $x_2 = c_{2t+1}$, $p_1 = 1$, $p_2 = \frac{1}{1+r_{t+1}}$)

Appendix 1: Why do we avoid corner solutions? (1 of 3)

1. Mathematical Pain

Interior: Euler Equation works.

$$u'(c_1) = \beta(1+r)u'(c_2)$$

Corner: Equality fails. Must use **Kuhn-Tucker** conditions. The algebra becomes piecewise and messy.

2. Economic Logic

Corner solutions often imply absurd outcomes, like $c_1 = 0$ (starvation).

The **Inada Condition** fixes this by making the first unit of consumption infinitely valuable (survival is priority).

Appendix 1: Why do we avoid corner solutions? (2 of 3)

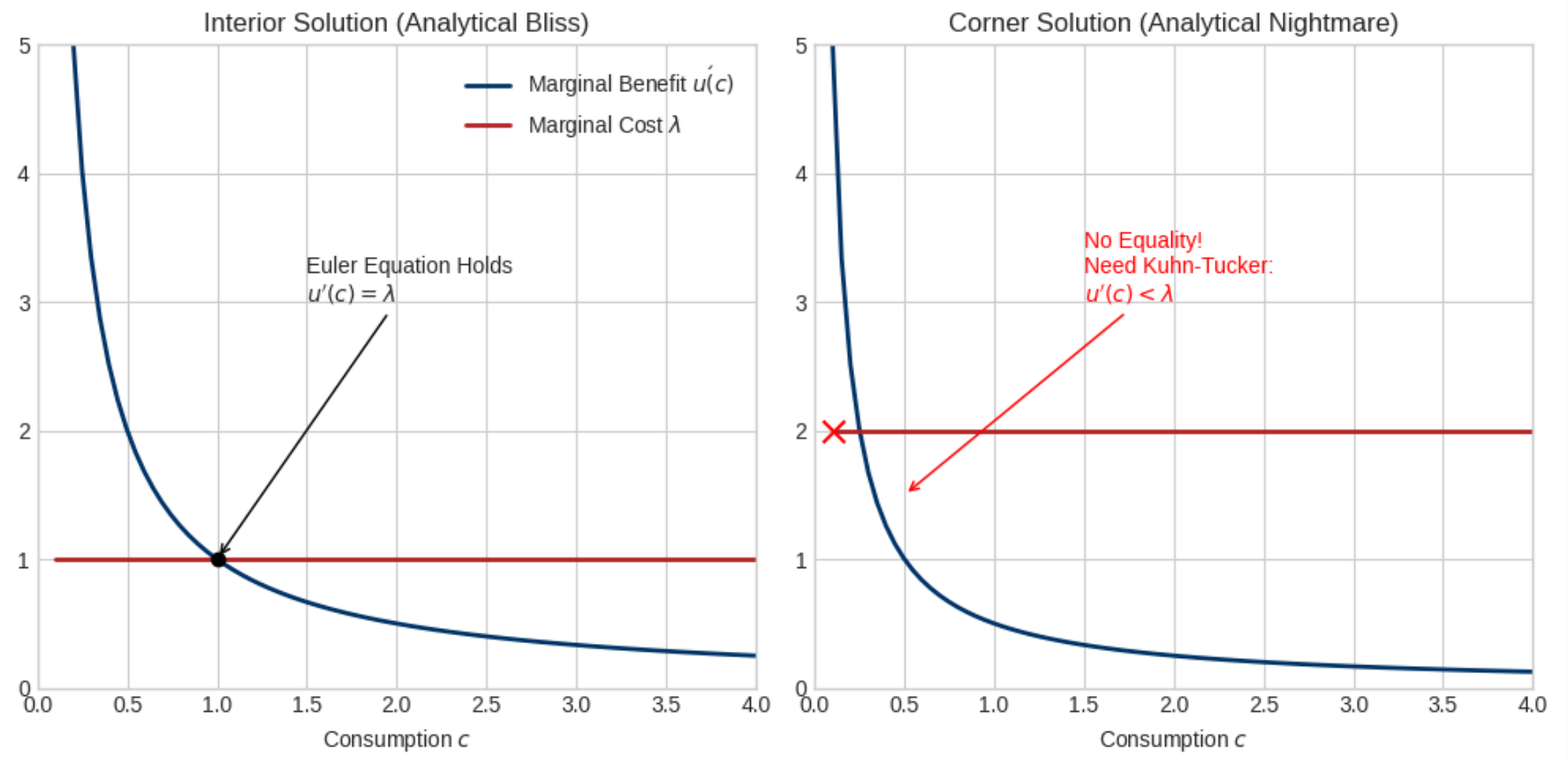
3. Dynamic Stability

Interior: Smooth capital paths.

Corner: Can create kinks, non-differentiability, or even non-existence of equilibrium.

Inada condition ensures existence, uniqueness, and smoothness.

Appendix 1: Why do we avoid corner solutions? (3 of 3)



Appendix 2: The Kuhn-Tucker Conditions: An Intuitive Guide

The Problem: Maximize $f(x)$ subject to $g(x) \leq c$.

The Lagrangian:

$$\mathcal{L} = f(x) + \lambda[c - g(x)]$$

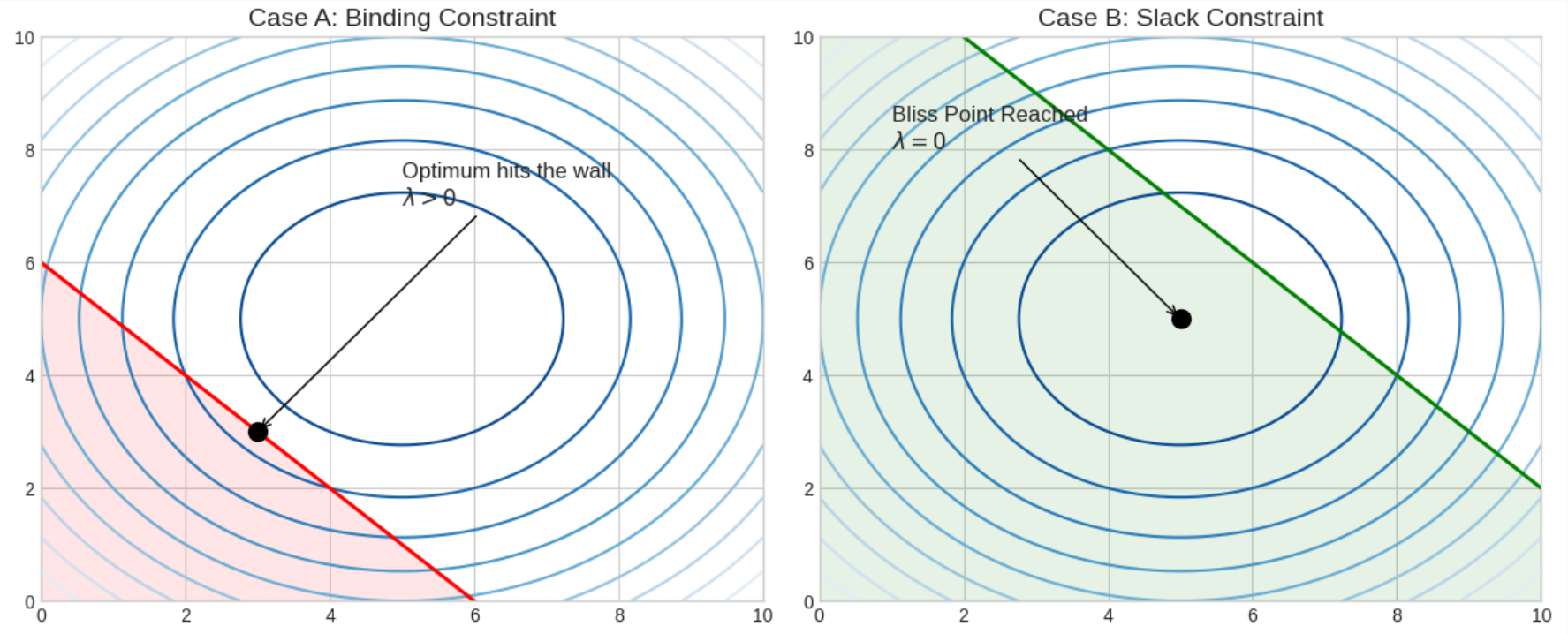
Here, λ is the **shadow price** of the constraint.

Complementary Slackness

$$\lambda \cdot [c - g(x)] = 0$$

Translation: Either the constraint binds ($g(x) = c$) OR the price is zero ($\lambda = 0$). *You cannot have a price for something you don't use fully.*

Appendix 2: Binding constraint vs. Slack constraint



- **Case A (Binding):** The wall stops you. The wall has value ($\lambda > 0$).
- **Case B (Slack):** The wall is far away. It is worthless ($\lambda = 0$).

Appendix 3

Prove that $\lim_{\theta \to 1} \frac{c^{1-\theta}-1}{1-\theta} = ln(c)$

Proof.
Note that $\lim_{\theta \to 1}(c^{1-\theta} - 1) = 1 - 1 = 0$, $\lim_{\theta \to 1}(1-\theta) = 0$. Therefore the limit is of "$\frac{0}{0}$" type. We need to use the L'Hôspital's rule to solve it. That is, we differentiate the numerator and the denominator with respect to θ separately, then compute the limit of the ratio of the two derivatives. We first do a bit of transformation for the limit:

$$\lim_{\theta \to 1} \frac{c^{1-\theta} - 1}{1-\theta} = \lim_{\theta \to 1} \frac{e^{ln(c^{1-\theta})} - 1}{1-\theta} = \lim_{\theta \to 1} \frac{e^{(1-\theta)ln(c)} - 1}{1-\theta}$$

Apply L'Hôspital's rule:

$$\lim_{\theta \to 1} \frac{e^{(1-\theta)ln(c)} - 1}{1-\theta} = \lim_{\theta \to 1} \frac{e^{(1-\theta)ln(c)}[-ln(c)]}{-1} = ln(c)$$

$\square$

Appendix 4

Show $\frac{1}{\theta}$ in the CRRA utility function is the elasticity of substitution between consumption in the two periods.

Solution: Since the first derivative of the CRRA utility function is

$$u'(c) = c^{-\theta}$$

then, the marginal rate of substitution (MRS) is:

$$\frac{u'(c_1)}{u'(c_2)} = \frac{c_1^{-\theta}}{c_2^{-\theta}} = (\frac{c_2}{c_1})^{\theta}$$

solving for $\frac{c_2}{c_1}$, get:

$$\frac{c_2}{c_1} = (\frac{u'(c_1)}{u'(c_2)})^{\frac{1}{\theta}}$$

Hence, $\frac{1}{\theta}$ is the elasticity of the ratio of consumption in the two periods with respect to MRS. By definition, $\frac{1}{\theta}$ is then the elasticity of substitution, which is constant for the CRRA utility function.

Appendix 4: Elasticity of Intertemporal Substitution (EIS) (1 of 2)

Definition

The elasticity of intertemporal substitution (EIS) is defined as:

$$\text{EIS} = \frac{d\ln(c_2/c_1)}{d\ln(\text{MRS})}$$

From CRRA utility:

Recall that

$$\frac{c_2}{c_1} = \left(\frac{u'(c_1)}{u'(c_2)}\right)^{\frac{1}{\theta}}$$

Taking logs:

$$\ln\left(\frac{c_2}{c_1}\right) = \frac{1}{\theta}\ln\left(\frac{u'(c_1)}{u'(c_2)}\right)$$

Appendix 4: Elasticity of Intertemporal Substitution (EIS) (2 of 2)

Differentiate both sides:

$$\frac{d\ln(c_2/c_1)}{d\ln(\text{MRS})} = \frac{1}{\theta}$$

Key Result

$$\boxed{\text{EIS} = \frac{1}{\theta}}$$

Because θ is constant in CRRA utility, the elasticity of substitution is constant.

Appendix 5: Derivation of Euler's Theorem under CRS (1 of 2)

1. Define Constant Returns to Scale (CRS)

A production function $Y = F(K, L)$ exhibits CRS if it is homogeneous of degree 1:

$$F(zK, zL) = zF(K, L), \quad \forall z > 0.$$

2. Differentiate with Respect to z

Differentiate both sides using the chain rule:

$$\frac{d}{dz} F(zK, zL) = \frac{d}{dz}[zF(K, L)].$$

$$\frac{\partial F(zK, zL)}{\partial (zK)} \cdot K + \frac{\partial F(zK, zL)}{\partial (zL)} \cdot L = F(K, L).$$

Appendix 5: Derivation of Euler's Theorem under CRS (2 of 2)

3. Evaluate at $z = 1$

Setting $z = 1$:

$$\frac{\partial F(K, L)}{\partial K} \cdot K + \frac{\partial F(K, L)}{\partial L} \cdot L = F(K, L).$$

4. Final Equation (Euler's Theorem)

Using marginal products:

$$MPK = \frac{\partial F}{\partial K}, \qquad MPL = \frac{\partial F}{\partial L},$$

$$\boxed{Y = MPK \cdot K + MPL \cdot L}$$

Core idea: Under CRS and perfect competition, aggregation is exact.

Step 1: Suppose there are N identical firms

Each firm uses (K_i, L_i).

$$K = \sum_i K_i, \qquad L = \sum_i L_i$$

Total output:

$$Y = \sum_i F(K_i, AL_i)$$

Step 2: Allocate inputs proportionally

Let

$$K_i = \frac{K}{N}, \qquad L_i = \frac{L}{N}$$

Then:

$$Y = N \cdot F\left(\frac{K}{N}, A\frac{L}{N}\right)$$

Step 3: Use CRS

Since F is homogeneous of degree 1:

$$F\left(\frac{K}{N}, A\frac{L}{N}\right) = \frac{1}{N}F(K, AL)$$

Therefore:

$$Y = F(K, AL)$$

Key Insight

Aggregate output depends only on total inputs, not on how many firms you split them across.

Conclusion: We can set $N = 1$ WLOG (without loss of generality).

World A: Entry *matters*

In models with:

- increasing returns,
- fixed costs,
- monopolistic competition,
- free entry,

the **number of firms** affects:

- markups and profits,
- product variety,
- measured productivity,
- wages and welfare.

Implication: you must model entry/exit explicitly.

World B: Entry is *irrelevant*

With **CRS + perfect competition**:

- profits are zero: $\pi = 0$,
- no fixed cost $\Rightarrow$ no entry condition,
- firm count does not change aggregates ($Y = F(K, AL)$).

Implication: tracking entry adds no macro content.

Bottom Line

In neoclassical growth, we can set the number of firms to 1 WLOG and focus on **capital accumulation, technology, and households' saving decisions**.

Appendix 8

Prove that $\frac{\partial F(K_t, A_t L_t^D)}{\partial K_t} = f'(k_t)$ and $\frac{\partial F(K_t, A_t L_t^D)}{\partial A_t L_t^D} = f(k_t) - f'(k_t)k_t$

Proof.

Recall that $F(K_t, A_t L_t^D) = A_t L_t^D f(k_t)$, and $k_t = \frac{K_t}{A_t L_t^D}$, therefore:

$$\frac{\partial F(K_t, A_t L_t^D)}{\partial K_t} = A_t L_t^D f'(k_t) \frac{\partial k_t}{\partial K_t} = A_t L_t^D f'(k_t) \frac{1}{A_t L_t^D} = f'(k_t)$$

$$\begin{aligned}
\frac{\partial F(K_t, A_t L_t^D)}{\partial A_t L_t^D} &= f(k_t) + A_t L_t^D f'(k_t) \frac{\partial k_t}{\partial A_t L_t^D} \\
&= f(k_t) + A_t L_t^D f'(k_t)[-\frac{K_t}{(A_t L_t^D)^2}] \\
&= f(k_t) - f'(k_t) \frac{K_t}{A_t L_t^D} \\
&= f(k_t) - f'(k_t) k_t
\end{aligned}$$

□

Appendix 9 (1 of 2)

Find the Euler equation for the following problem:

$$\max_{c_{1t}, c_{2t+1}} U(c_{1t}, c_{2t+1}) = u(c_{1t}) + \beta u(c_{2t+1})$$

$$s.t. \quad c_{1t} + \frac{1}{1 + r_{t+1}} c_{2t+1} = A_t w_t$$

Solution: Set up the Lagrangian function:

$$\mathcal{L} = U(c_{1t}, c_{2t+1}) + \lambda(A_t w_t - c_{1t} - \frac{1}{1 + r_{t+1}} c_{2t+1})$$

where λ is the Lagrangian multiplier. FOCs:

$$\frac{\partial \mathcal{L}}{\partial c_{1t}} = U_1(c_{1t}, c_{2t+1}) - \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial c_{2t+1}} = U_2(c_{1t}, c_{2t+1}) - \lambda \frac{1}{1 + r_{t+1}} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = A_t w_t - c_{1t} - \frac{1}{1 + r_{t+1}} c_{2t+1} = 0$$

Appendix 9 (2 of 2)

Then, we have:

$$U_1(c_{1t}, c_{2t+1}) = \lambda$$
$$U_2(c_{1t}, c_{2t+1}) = \lambda \frac{1}{1 + r_{t+1}}$$

It is apparently that:

$$\frac{U_1(c_{1t}, c_{2t+1})}{U_2(c_{1t}, c_{2t+1})} = 1 + r_{t+1}$$

This is the consumption Euler equation. Together with the lifetime budget constraint, you get 2 equations with two unknowns (c_{1t} and c_{2t+1}). Then, the individual's problem is solved.